

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy & material cycles: natural vs human systems - Set 1

Mark scheme

AQA spec 3.6.2-3.6.3; Sustainability; textbook pp. 373-374

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 4 points below. 1 Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis – human processes usually use high energy-density non-renewable fossil fuels 1 Natural processes occur at low temperatures because enzymes lower the activation energy of reactions – human processes need high temperatures and pressures 1 Natural processes produce no toxic wastes / their products are raw materials for other processes – industrial processes emit CO₂ and NO_x 1 So natural fixation is sustainable indefinitely with a low carbon footprint Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • sustainable → meeting present needs without stopping future generations meeting theirs Do not credit • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4

Question	Marking guidance	Marks
01.2	Any 3 of 5 1 mark each for any 3 of the 5 points below, in any order, maximum 3. 1 Hydrological cycle (evaporation, precipitation) 1 Carbon cycle / photosynthesis 1 Atmospheric circulation / winds 1 Thermohaline circulation (density differences from heating / evaporation) 1 Decomposition Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Thermohaline → the ocean conveyor driven by temperature and salinity Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	3
01.3	1 mark each 1 mark for each of the 3 points below. 1 Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting 1 Anaerobic digestion and composting treat wastes – enzymes in industry (biological washing powders, biofuels from algae / microorganisms) 1 Heat pumps and solar thermal capture low-density heat – low-temperature manufacturing using biological polymers – fuel cells instead of combustion Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
02.1	1 mark each 1 mark for each of the 4 points below. 1 Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins) – wastes of one process are raw materials for another so nothing is depleted or accumulates 1 Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials 1 Human systems are linear: extract - process - manufacture - use - discard – minerals are dispersed after use or mixed so recovery is difficult 1 Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up – fossil fuel use is inherently linear and unsustainable Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. biodegradable → broken down by decomposers persistent → does not break down / stays in the environment for a long time Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Question	Marking guidance	Marks
02.2	1 mark each 1 mark for each of the 2 points below. 1 Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life 1 They leach from landfill or are released by incineration – persistent, bioaccumulating Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. sustainability → meeting present needs without stopping future generations meeting theirs incineration → burning waste, usually with energy recovery persistent → does not break down / stays in the environment for a long time Do not credit 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
02.3	1 mark each 1 mark for each of the 4 points below. 1 Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) 1 Easy identification of materials (coding) and separation of components so they can be recycled 1 Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) 1 Use of recycled and recyclable / biodegradable materials – reduced packaging and material use (Cradle to Cradle) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. biodegradable → broken down by decomposers	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
03.1	Any 2 of 6 1 mark each for any 2 of the 6 points below, in any order, maximum 2. 1 Hydrological cycle 1 Carbon cycle 1 Nitrogen cycle 1 Atmospheric circulation 1 Thermohaline circulation 1 Photosynthesis Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Thermohaline → the ocean conveyor driven by temperature and salinity Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	2

Question	Marking guidance	Marks
03.2	1 mark each 1 mark for each of the 2 points below. 1 Enzymes reduce the activation energy of reactions 1 So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. nitrogen fixation → nitrogen gas converted into ammonium or other nitrogen compounds	2
03.3	Any 1 of 3 1 mark each for any 1 of the 3 points below, in any order, maximum 1. 1 Haber process (nitrogen fixation) 1 Incineration of wastes (decomposition) 1 Many manufacturing processes Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. nitrogen fixation → nitrogen gas converted into ammonium or other nitrogen compounds Incineration → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
04.1	1 mark each 1 mark for each of the 4 points below. 1 Energy: low energy-density renewable (solar) / non-renewable fossil fuels 1 Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion 1 Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate 1 Temperature: low / high (Haber process, incineration) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - The converse of any point, ie the same comparison written the other way round. biodegradable → broken down by decomposers persistent → does not break down / stays in the environment for a long time incineration → burning waste, usually with energy recovery Do not credit 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
04.2	1 mark each 1 mark for each of the 2 points below. 1 Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) 1 They are re-used in cycles so there is no depletion of raw materials or build-up of wastes Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
04.3	2 marks 2 marks for the point below; award fewer where it is only partly made. 2 Use often disperses the minerals (eg in products spread over the world / corrosion) OR produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. sustainable → meeting present needs without stopping future generations meeting theirs Do not credit 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
04.4	1 mark each 1 mark for each of the 2 points below. 1 They cannot easily be removed from end-of-life appliances and are not biodegradable 1 Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. biodegradable → broken down by decomposers persistent → does not break down / stays in the environment for a long time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
05.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 C: Waste products that are the raw materials for other processes	1
05.2	1 mark each 1 mark for each of the 2 points below. 1 Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) 1 The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂ Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. nitrogen fixation → nitrogen gas converted into ammonium or other nitrogen compounds sustainable → meeting present needs without stopping future generations meeting theirs Do not credit 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Question	Marking guidance	Marks
05.3	1 mark each 1 mark for each of the 2 points below. 1 Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials 1 Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • incineration → burning waste, usually with energy recovery • sustainable → meeting present needs without stopping future generations meeting theirs Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	